# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME — SET A

Chapter 14: **Math of Space: Surface Area and Volume** | Class 9 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 158 cm$^2$ | $2(lw+wh+lh)=2(40+15+24)=158$ |
| 2 | **(d)** 210 cm$^3$ | $lwh=7\times6\times5=210$ |
| 3 | **(d)** 96 cm$^2$ | side $=4$ cm, TSA $=6a^2=96$ |
| 4 | **(a)** 528 cm$^2$ | $2\pi rh=528$ |
| 5 | **(b)** 550 cm$^2$ | $\pi rl=550$ |
| 6 | **(b)** 10 cm | $l=\sqrt{6^2+8^2}=10$ |
| 7 | **(a)** 38808 cm$^3$ | $\frac{4}{3}\pi r^3=38808$ |
| 8 | **(d)** 19404 cm$^3$ | $\frac{2}{3}\pi r^3$ |
| 9 | **(b)** $\pi r(2h+r)$ | curved surface plus one base |
| 10 | **(a)** Both A and R are true and R is the correct explanation of A | experimental verification |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Side $=\sqrt[3]{343}=7$ cm (1). TSA $=6\times49=294$ cm$^2$ (1). | 1+1 |
| 12 | $V=2.5\times2\times1.2=6$ m$^3$ (1) $=6000$ litres (1). | 1+1 |
| 13 | CSA $=2\pi rh=2\times\frac{22}{7}\times7\times20$ (1) $=880$ cm$^2$ (1). | 1+1 |
| 14 | $l=\sqrt{r^2+h^2}=\sqrt{49+576}$ (1) $=25$ cm (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Original TSA $=6\times36=216$ cm$^2$ (1). Number of small cubes $=216$, each of TSA 6 cm$^2$, total $=1296$ cm$^2$ (1). Ratio $=216:1296=1:6$ (1). | 1+1+1 |
| 16 | Volume of godown $=4800$ m$^3$ (1). Volume of one box $=2.5$ m$^3$ (1). Number of boxes $=4800\div2.5=1920$ (1). | 1+1+1 |
| 17 | (i) corner cubes: 8 (1). (ii) 12 edges $\times$ 2 cubes $=24$ (1). (iii) 6 faces $\times$ 4 cubes $=24$ (½); (iv) inner cube $2\times2\times2=8$ (½). Check: $8+24+24+8=64$. | 1+1+½+½ |
| | **OR:** $V=\pi r^2h=\frac{22}{7}\times0.49\times1.5$ (1) $=2.31$ m$^3$ (1) $=2310$ litres (1). | 1+1+1 |
| 18 | $2\pi rh=264\Rightarrow r=\frac{264}{2\times\frac{22}{7}\times7}=6$ cm (1). $V=\pi r^2h=\frac{22}{7}\times36\times7$ (1) $=792$ cm$^3$ (1). | 1+1+1 |
| | **OR:** (i) $V_1:V_2=r_1^2h_1:r_2^2h_2=(1\times4):(4\times3)$ (1) $=1:3$ (1). (ii) $\text{CSA}_1:\text{CSA}_2=r_1h_1:r_2h_2=4:6=2:3$ (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Cut the curved surface along a line parallel to the axis and open it out: it is a rectangle with sides $2\pi r$ and $h$, so CSA $=2\pi rh$ (2). Adding two circular ends, TSA $=2\pi rh+2\pi r^2=2\pi r(h+r)$ (1). For $r=7$, $h=15$: CSA $=660$ cm$^2$ (1); TSA $=660+308=968$ cm$^2$ (1). | 2+1+1+1 |
| | **OR:** (i) $V=\frac{22}{7}\times1.96\times3=18.48$ m$^3=18{,}480$ litres (1½). (ii) CSA $=2\times\frac{22}{7}\times1.4\times3=26.4$ m$^2$ (1). (iii) Cost $=26.4\times25=$ ₹660 (1). (iv) $18480\div120=154$ days (1½). | 1½+1+1+1½ |
| 20 | (i) $l=\sqrt{49+576}=25$ cm (1). (ii) CSA $=\pi rl=550$ cm$^2$ (1); TSA $=550+154=704$ cm$^2$ (1). (iii) $V=\frac{1}{3}\pi r^2h=1232$ cm$^3$ (1). (iv) The cylinder has volume $3696$ cm$^3$, three times the cone (1). | 1+2+1+1 |
| | **OR:** (i) $l=25$ m (1). (ii) Canvas $=\pi rl=550$ m$^2$ (1½). (iii) $V=\frac{1}{3}\pi r^2h=1232$ m$^3$ (1½). (iv) $1232\div11=112$ people (1). | 1+1½+1½+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET B

Chapter 14: **Math of Space: Surface Area and Volume** | Class 9 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 248 cm$^2$ | $2(lw+wh+lh)=2(60+24+40)=248$ |
| 2 | **(a)** 600 cm$^3$ | $lwh=12\times10\times5=600$ |
| 3 | **(a)** 150 cm$^2$ | side $=5$ cm, TSA $=6a^2=150$ |
| 4 | **(a)** 1540 cm$^3$ | $\pi r^2h=1540$ |
| 5 | **(d)** 660 cm$^2$ | $\pi rl=660$ |
| 6 | **(c)** 13 cm | $l=\sqrt{5^2+12^2}=13$ |
| 7 | **(b)** 4851 cm$^3$ | $\frac{4}{3}\pi r^3=4851$ |
| 8 | **(a)** 120 cm$^3$ | $\frac{1}{3}\times36\times10=120$ |
| 9 | **(d)** 8 times | $V\propto r^3$ |
| 10 | **(d)** A is false but R is true | SA becomes four times; A is false |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $V=\frac{2}{3}\pi r^3=\frac{2}{3}\times\frac{22}{7}\times10.5^3$ (1) $=2425.5$ cm$^3$ (1). | 1+1 |
| 12 | Surface area $=4\pi r^2$ (1); volume $=\frac{4}{3}\pi r^3$ (1). | 1+1 |
| 13 | A cone has one-third the volume of the cylinder with the same base and height (1). Fill the cone with salt and empty it into the cylinder; it takes exactly three fills (1). | 1+1 |
| 14 | Volume of cuboid $=480$ cm$^3$; volume of one cube $=8$ cm$^3$ (1). Number $=60$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Volume of sphere $=\frac{4}{3}\pi(27)=36\pi$ cm$^3$ (1). Volume of one cone $=\frac{1}{3}\pi(1)(3)=\pi$ cm$^3$ (1). Number $=36\pi\div\pi=36$ (1). | 1+1+1 |
| 16 | $l=\sqrt{49+576}=25$ m (1). Canvas $=\pi rl=\frac{22}{7}\times7\times25=550$ m$^2$ (1). Cost $=550\times120=$ ₹66,000 (1). | 1+1+1 |
| 17 | $\pi rl=462\Rightarrow r=\frac{462}{\frac{22}{7}\times21}=7$ cm (1). Base area $=\frac{22}{7}\times49=154$ cm$^2$ (1). TSA $=462+154=616$ cm$^2$ (1). | 1+1+1 |
|  | **OR:** The solid is a cone with $r=7$, $h=24$, $l=25$ (1). $V=\frac{1}{3}\times\frac{22}{7}\times49\times24=1232$ cm$^3$ (1). CSA $=\pi rl=550$ cm$^2$ (1). | 1+1+1 |
| 18 | Water forms a similar cone of height $\frac{1}{3}$ the cup's height (1), so its radius is also $\frac{1}{3}$ (1). Volume ratio $=\left(\frac{1}{3}\right)^3=\frac{1}{27}$ (1). | 1+1+1 |
|  | **OR:** $V=\frac{1}{3}\times81\times10=270$ cm$^3$ (2). Prism volume $=810$ cm$^3$, so the pyramid is one-third of it (1). | 2+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) A cylinder of radius $r$ and height $2r$ fits tightly over the sphere; its curved surface area is $2\pi r\times2r=4\pi r^2$, which equals the sphere's surface area (2). (ii) Surface area $=4\pi r^2=4\times\frac{22}{7}\times110.25=1386$ cm$^2$ (1½). Volume $=\frac{4}{3}\pi r^3=4851$ cm$^3$ (1½). | 2+1½+1½ |
|  | **OR:** (i) 21 cm $\times$ 7 cm $\times$ 7 cm (1). (ii) $21\times7\times7=1029$ cm$^3$ (1). (iii) $3\times\frac{4}{3}\times\frac{22}{7}\times3.5^3=539$ cm$^3$ (2). (iv) $1029-539=490$ cm$^3$ (1). | 1+1+2+1 |
| 20 | (i) $l=\sqrt{10.5^2+14^2}=\sqrt{306.25}=17.5$ cm (1). (ii) Cone CSA $=\pi rl=\frac{22}{7}\times10.5\times17.5=577.5$ cm$^2$ (1); hemisphere CSA $=2\pi r^2=693$ cm$^2$ (1); total $=1270.5$ cm$^2$ (1). (iii) $V=1617+2425.5=4042.5$ cm$^3$ (1). | 1+3+1 |
|  | **OR:** (i) $V=\pi r^2h=\frac{22}{7}\times144\times35=15{,}840$ cm$^3$ (1). (ii) $\frac{4}{3}\pi r^3=38.808$ cm$^3$ (1). (iii) 50% of $38.808=19.404$ cm$^3$ (1). (iv) $15840\div19.404\approx816$ golgappas (1). (v) The result depends on the estimated size and the fraction of pani; guesstimates give a range, not an exact value (1). | 1+1+1+1+1 |

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